FINOS: An Agent-Orchestrated Financial Intelligence System

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***Abstract*—Financial dashboards usually show budgets, transactions, risk indicators and forecasts in separate views, leaving the analyst to relate them and to judge whether a number deserves trust. This paper describes FINOS (Financial Intelligence and Operations System), an academic prototype that places financial monitoring, budget management, risk analysis, cash-flow forecasting and decision support in one web platform. Seven algorithm families (A* search, constraint satisfaction with backtracking, a rules engine, Bayesian risk estimation, minimax, Isolation Forest, and ordinary least squares with exponential smoothing) are coordinated by 11 specialized agents through a nine-stage workflow: observe, analyze, reason, plan, validate, simulate, verify, audit and replan. We explain what each method can and cannot support, why feasibility is not optimality, and why an anomaly is not fraud. The project team reports that 20 of 20 automated tests passed; we present these as team-provided results that we have not reproduced. No forecasting accuracy, detection quality or financial benefit is claimed, and we specify the evidence needed to evaluate FINOS scientifically.**

***Index Terms*—financial decision support, multi-agent systems, budget constraints, risk estimation, anomaly detection, cash-flow forecasting, Isolation Forest, exponential smoothing.**

I. Introduction

Financial decision-making needs more than displayed numbers. An analyst must relate budget limits to expenditure, recognize unusual changes, judge risk and anticipate how present conditions affect future cash flows. Conventional tools spread these concerns over separate dashboards, so the overall picture is assembled by hand. A chart-only system shows *what* is happening, but not why a pattern matters, which constraints bind, or how alternative actions compare.

FINOS addresses this gap by connecting financial data, analytical algorithms, specialized software agents, validation rules, scenario simulation and reviewable recommendations in one process (Fig. 1). The agent view follows the standard characterization of agents as autonomous, socially able, reactive and proactive entities [1]. Because outputs may influence financial decisions, the design stresses validation and human review, consistent with supervisory model-risk guidance [2] and the human-oversight emphasis of the NIST AI Risk Management Framework [3].

Scope. FINOS is an academic prototype. Financial activity, market ticks and approvals may be simulated, and it does not execute real transfers or live securities trades. We separate information from the project description, external scholarly background and proposed future evaluation. No experimental measurements, accuracy scores or deployment validation are reported, because none were supplied.

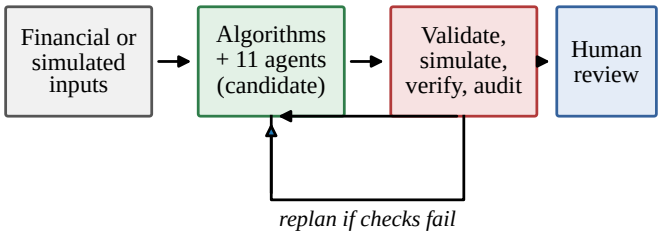


Figure 1. FINOS decision-support concept: algorithms and agents generate a candidate, which is accepted only after validation, simulation, verification and audit, and is then reviewed by a human.

Research question. How can multiple financial-analysis algorithms and specialized software agents be integrated into a structured decision-support system that improves the organization, interpretation, validation and review of financial information?

This paper makes an integration and prototype contribution: (i) a unified financial decision-support design with centralized monitoring; (ii) a mapping of seven algorithm families to distinct analytical tasks, with their equations and cautions; (iii) a nine-stage workflow that separates generating a candidate from accepting it; (iv) a modular web and backend architecture; and (v) an evaluation protocol with team-reported test results. It does not claim a novel algorithm, superior accuracy, improved institutional efficiency or measurable savings.

The remainder of the paper is organized as follows. Section II reviews related work, Section III formulates the problem, Section IV presents the method, Section V describes the implementation, Section VI covers testing, Section VII discusses the findings, Section VIII covers limitations and ethics, and Section IX concludes.

II. Related Work

A. Agents and Multi-Agent Systems

Wooldridge and Jennings distinguish agent theory, architectures and languages [1], and Wooldridge's textbook treats coordination and task decomposition among agents [4]. ReAct showed that interleaving reasoning traces with actions helps a language-model agent track plans and handle exceptions, with improved interpretability over baselines without traces [5]. FINOS borrows the idea of structured observe–reason–act cycles, but according to

the project description its methods are classical, deterministic or statistical rather than language-model based.

B. Anomaly Detection

Chandola et al. survey anomaly detection and stress that every technique rests on assumptions about what distinguishes normal from anomalous behavior [6]. This is why FINOS treats an anomaly score as a review signal and not as a finding of misconduct.

C. Forecasting and Its Evaluation

Hyndman and Athanasopoulos give a standard treatment of regression and exponential smoothing [7]. Hyndman and Koehler compare accuracy measures and recommend scaled errors for comparisons across series [8]. Bergmeir and Benítez examine cross-validation for time-series predictor evaluation [9]. These works define the standard FINOS forecasts would have to meet.

D. Model Governance and Positioning

Model-risk guidance emphasizes validation, documentation and effective challenge [2], and the NIST framework addresses trustworthiness and human oversight [3]. These motivate the validate, verify and audit stages. We did not conduct a systematic survey of competing financial platforms, so no comparative claim is made.

III. Problem Formulation

A. Decision-Support Setting

Let O_t denote the financial or simulated observations available at time t, $\mathscr{C}$ a set of configured constraints, $\mathscr{R}$ a set of configured rules, and a a candidate action or recommendation produced by an algorithm. We formalize acceptance (our notation, not taken from the FINOS code) as

$$\text{accept}(a) = \bigwedge_{c \in \mathscr{C}} c(a) \;\wedge\; \bigwedge_{r \in \mathscr{R}} r(a) \;\wedge\; \text{verify}\big(\text{sim}(a, O_t)\big), \quad (1)$$

where sim runs a simulated scenario and verify checks internal consistency. A candidate that fails (1) is replanned; one that passes is still only a recommendation for human review.

Feasibility versus optimality. For a budget with total B, line allocations x_i and configured limits ℓ_i, u_i, a scenario is feasible if

$$\sum_{i=1}^{k} x_i = B, \qquad \ell_i \le x_i \le u_i. \quad (2)$$

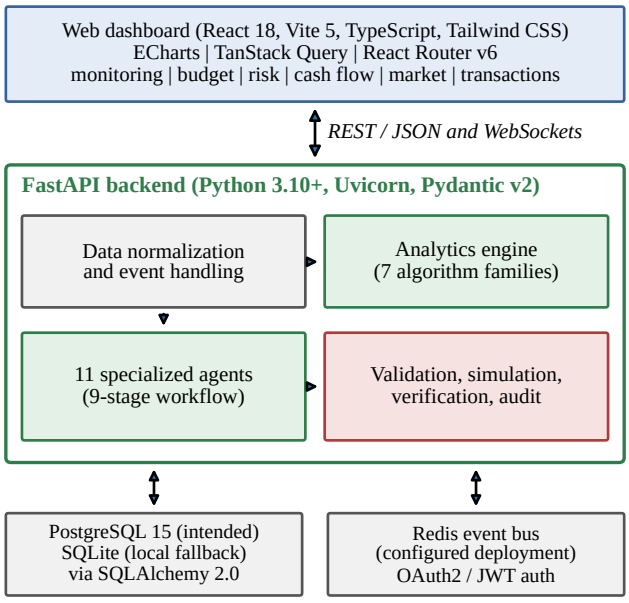
Satisfying (2) shows that limits are respected; it does not show that the allocation is economically or institutionally optimal, which would need an explicit objective.

B. System Requirements

Table I lists the requirements derived from the project objectives. They are design intentions; whether each is fully met in the code is not established here.

Table I
System requirements derived from the objectives

ID	Requirement	Addressed by
R1	Consolidated view of indicators, budgets, transactions, risk, cash flow	Dashboard
R2	Deterministic budget and limit checks	Rules, CSP
R3	Risk updating; flagging unusual records	Bayesian, IForest
R4	Forecasts of cash-flow trends	OLS, smoothing
R5	Candidates checked before display	Plan, Validate
R6	Scenario simulation, consistency checks	Simulate, Verify
R7	Recorded inputs, methods, outputs	Audit
R8	Revision on failed checks	Replan
R9	Authenticated access	OAuth2/JWT
R10	Human review of outputs	Review stages



Intended architecture from the project description

Figure 2. Intended system architecture (project description). Agent-to-agent communication is not shown because it is not specified.

IV. Proposed Method

A. Architecture

Fig. 2 shows the intended architecture from the project description. Financial or simulated inputs are normalized, delivered as events, analyzed by the algorithm families and agents, validated and simulated, persisted through application services, and shown in the React dashboard for user review. This is the intended design; we do not claim that every component was independently audited or that every connection was verified in a live production environment.

B. Nine-Stage Agent Workflow

The project description states that 11 specialized agents are coordinated through nine stages (Fig. 3). **The names, responsibilities, communication protocols and autonomy of the agents are not specified in the supplied material and are therefore not described**; they must be confirmed from source code (**[AGENT TABLE TO BE COMPLETED]**).

Algorithm 1 Orchestration loop (conceptual)

```
O ← Observe(); F ← Analyze(O)
for i = 1 to B_max do
    z ← Reason(F); a ← Plan(z)
    if Validate(a) and Verify(Simulate(a)) then
        Audit(a); return a for human review
    end if
    F ← Replan(F, failed checks, new data)
end for
Audit(failure); return no candidate
```

The workflow gives heterogeneous methods a shared contract. Each method contributes at the analyze, reason or plan stages; every candidate then passes through the same validate, simulate and verify gates before it is recorded and shown. This separates *generating* from *accepting* a candidate. Algorithm 1 gives the conceptual loop; the mapping of methods to stages is our proposal for explanation, and the iteration bound $B_{\max}$ is not stated in the description.

Why human review remains necessary. Passing checks shows conformity to configured rules, not that the rules reflect policy or that the recommendation is wise. Model-risk guidance calls for independent validation and effective challenge [2], and AI risk guidance stresses human oversight [3]. In FINOS any simulated approval or dispatch is a prototype demonstration, not a legally effective approval.

C. The Seven Algorithm Families

Each paragraph gives the standard definition (external background) and then the intended role in FINOS (project description).

1) A* search. A* expands the frontier node with the lowest estimated total cost [10, 11]:

$$f(n) = g(n) + h(n), \tag{3}$$

where $g(n)$ is the accumulated cost and $h(n)$ estimates the remaining cost. If h never overestimates the true remaining cost it is admissible, and A* returns a minimum-cost solution; consistency additionally requires $h(n) \le c(n,n') + h(n')$ [10, 11]. In FINOS, A* can support a planning problem once states, actions, costs and a heuristic are defined; the concrete use case must be confirmed from source (**[A* USE CASE]**). Optimality holds only relative to the cost model and heuristic.

2) CSP and backtracking. A constraint satisfaction problem is a triple (X, D, C) of variables, domains and constraints [11–13]. Backtracking assigns variables one by one and returns to an earlier choice when a constraint is violated; node, arc and path consistency prune values to reduce thrashing [12]. Worst-case time is exponential. In FINOS it can explore budget or workflow scenarios satisfying totals, limits, eligibility or dependencies, as in (2).

3) Financial rules engine. A rules engine evaluates condition–action rules (IF condition THEN outcome); efficient matching, for example the Rete algorithm [14], is well studied, though we do not claim FINOS uses it. FINOS uses rules for threshold checks, budget-limit checks, required-field validation and prerequisites. Rules are transparent and repeatable but only as correct as their encoding of policy, and they cannot detect patterns they were not written for.

4) Bayesian risk estimation. Bayes' theorem updates the probability of hypothesis H given evidence E [15]:

$$P(H \mid E) = \frac{P(E \mid H)P(H)}{P(E)}, \qquad P(E) = P(E \mid H)P(H) + P(E \mid \neg H)P(\neg H). \tag{4}$$

$P(H)$ is the prior, $P(E \mid H)$ the likelihood and $P(H \mid E)$ the posterior; equivalently, posterior odds equal the likelihood ratio times prior odds (Fig. 4c). FINOS uses it to update a modeled risk estimate. The estimate is meaningful only if prior, likelihoods and independence assumptions are appropriate; no empirical calibration is claimed.

5) Minimax. In an adversarial tree one player maximizes and the other minimizes a utility [11]:

$$V(s) = \begin{cases} U(s) & \text{terminal} \\ \max_a V(\mathrm{res}(s,a)) & \text{MAX} \\ \min_a V(\mathrm{res}(s,a)) & \text{MIN.} \end{cases} \tag{5}$$

Alpha-beta pruning reduces examined nodes without changing the value [16]. In FINOS it can evaluate bounded strategic scenarios if states, actions, utilities and an opponent model are defined (**[MINIMAX USE CASE]**). It is *not* a general financial forecasting method.

6) Isolation Forest. An unsupervised ensemble isolates observations by random feature and split selection; anomalies tend to have shorter average path lengths [17]. For sample size n,

$$s(x,n) = 2^{-E[h(x)]/c(n)}, \quad c(n) = 2H(n-1) - \tfrac{2(n-1)}{n}, \tag{6}$$

where $h(x)$ is the path length and $H(i) \approx \ln i + 0.5772$; scores near 1 indicate likely anomalies and scores well below 0.5 normal points (Fig. 4a). FINOS uses it to flag unusual records. **A score is a signal for review, not proof of fraud, misconduct or loss**; a legitimate seasonal payment can score high [6].

7) OLS and exponential smoothing. OLS minimizes the sum of squared residuals [18]:

$$\hat{\beta} = \arg\min_{\beta} \sum_t (y_t - x_t^\top \beta)^2 = (X^\top X)^{-1} X^\top y, \tag{7}$$

which suits linear relationships when specification and error assumptions hold; serially correlated errors can make standard errors unreliable [7, 18]. Simple smoothing and Holt's linear-trend method are [7, 19]

$$\ell_t = \alpha y_t + (1-\alpha)\ell_{t-1}, \quad \hat{y}_{t+h|t} = \ell_t, \tag{8}$$

$$\ell_t = \alpha y_t + (1-\alpha)(\ell_{t-1} + b_{t-1}), \tag{9}$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)b_{t-1}, \quad \hat{y}_{t+h|t} = \ell_t + h\,b_t, \tag{10}$$

with $0 < \alpha < 1$ (Fig. 4b); seasonal variants add a seasonal component. FINOS uses them for cash-flow trend projection; which variant and regressors are implemented must be confirmed from code.

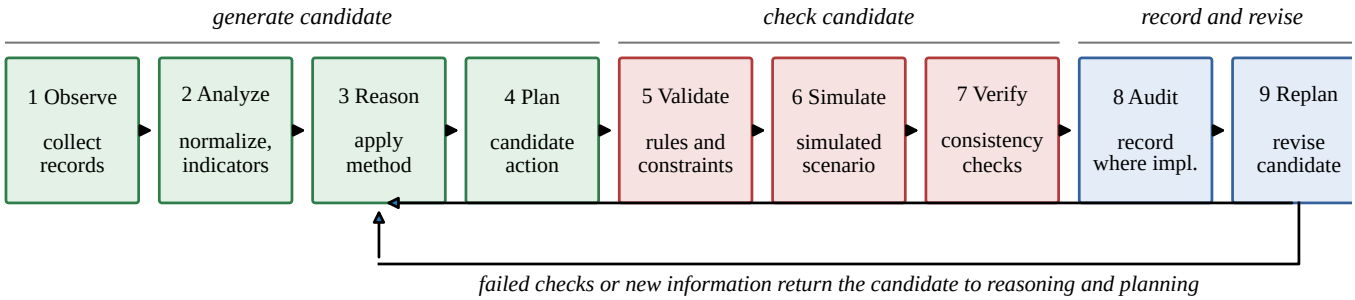


Figure 3. Nine-stage workflow. Stages 1–4 generate a candidate, stages 5–7 check it, stage 8 records and stage 9 revises it.

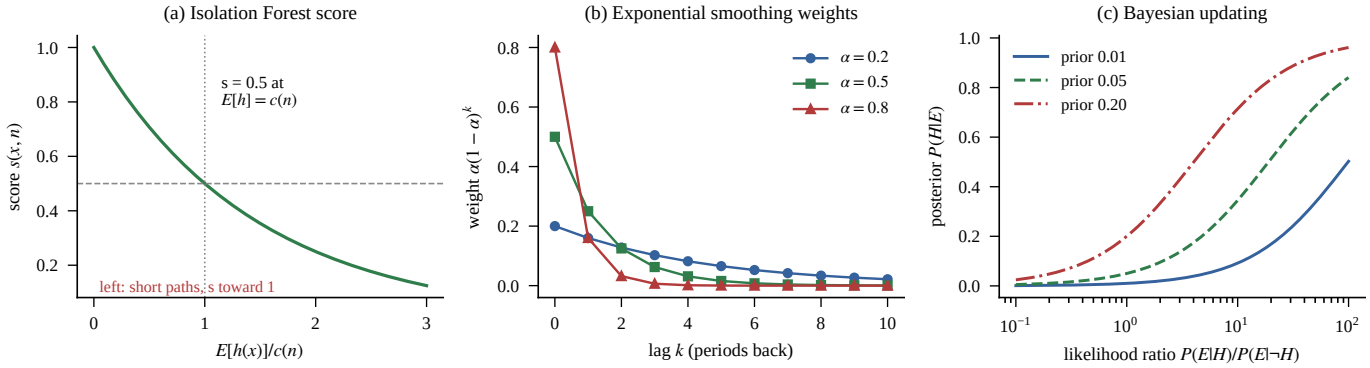


Figure 4. Illustrative curves computed directly from Eqs. (6), (8) and (4). They show the behavior of the formulas only and contain no FINOS data or results.

Table II
ALGORITHM FAMILIES, TASKS AND CAUTIONS

Family	Intended task	Key caution
A*	Planning/search	Needs admissible heuristic; use case unconfirmed
CSP	Budget, workflow feasibility	Feasible is not optimal
Rules	Policy checks	Rules may misstate policy
Bayesian	Update modeled risk	Not calibrated without evidence
Minimax	Bounded adversarial scenarios	Not a forecasting method
IForest	Unusual records	Anomaly is not fraud
OLS/ES	Cash-flow trends	Accuracy unmeasured

Table III
TECHNOLOGY STACK (PROJECT DESCRIPTION)

Layer	Technologies
Frontend	React 18, Vite 5, TypeScript, Tailwind CSS, Apache ECharts, Lucide icons, TanStack Query, React Router v6
Backend	Python 3.10+, FastAPI, SQLAlchemy 2.0, Pydantic v2, Uvicorn, WebSockets
Database	PostgreSQL 15 (intended); SQLite (local fallback)
Events	Redis event bus (configured deployment); WebSockets to browsers
Auth	OAuth2 with JWT (reported tested)

D. Financial-Analysis Modules

FINOS has six functional areas: (A) a monitoring dashboard consolidating indicators; (B) budget management for allocations, expenditure and limits; (C) risk analysis of configured indicators; (D) cash-flow forecasting; (E) market intelligence showing configured or *simulated* ticks, never a verified live feed; and (F) transaction and project workflows whose simulated approvals are demonstrations. Fig. 5 shows our proposed association of methods with modules; it is a conceptual reading to be confirmed from code. The modules interact: budget limits (B) are the constraints against which candidates are validated; forecasts (D) can be compared with budget capacity; risk and anomaly flags (C) derive from transaction records (F); and the dashboard (A) aggregates all of them.

V. SYSTEM IMPLEMENTATION

Table III lists the stack from the project description. It is the intended architecture; we make no claim of an independent audit.

FastAPI exposes the endpoints, Pydantic validates payloads, SQLAlchemy mediates database access and WebSockets can push updates where configured. The OAuth2/JWT scheme builds on the JSON Web Token [20] and OAuth 2.0 framework [21] standards; grant types, token lifetimes, signing algorithms and the role model are not stated and must be documented. Because SQLite and PostgreSQL may behave differently (concurrency, types), tests should state which was used. Repository structure, agent code, algorithm parameters, schemas, dataset provenance and the Redis channel design were not supplied and are not described.

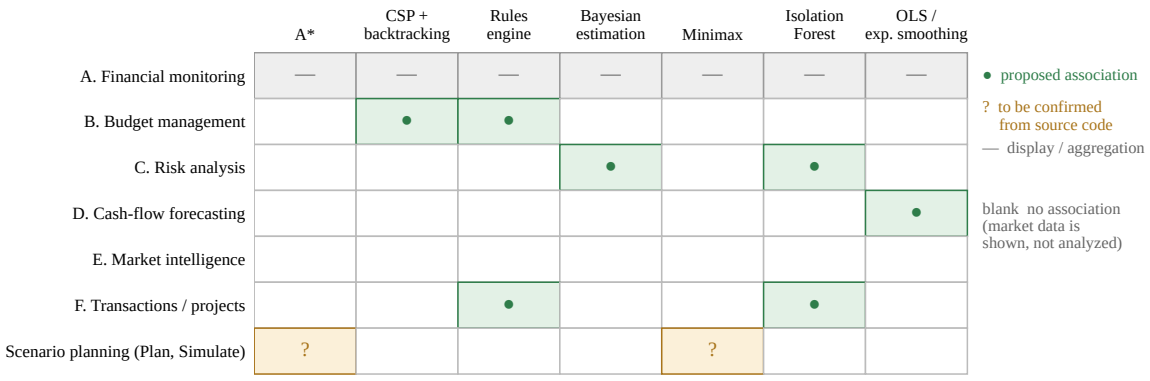


Figure 5. Proposed association of algorithm families with FINOS modules (conceptual; to be confirmed from source code).

Table IV
REPORTED AUTOMATED TESTS (TEAM-PROVIDED)

Category	Tests	Reported outcome
Algorithm	9	9 passed
Agent	2	2 passed
API	9	9 passed
Total	20	20 passed

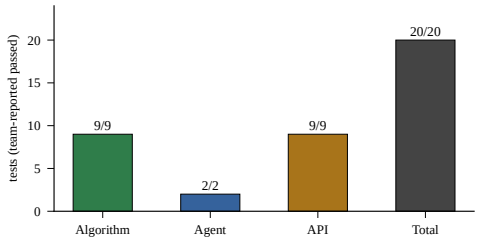


Figure 6. Team-reported test counts (not independently reproduced).

VI. TESTING METHODOLOGY AND RESULTS

A. Team-Reported Results

The team reports that **20 of 20 automated tests passed**: 9 algorithm, 2 agent and 9 API tests (Table IV, Fig. 6). The API tests cover health status, dashboard metrics, global pulse, transactions, budgets, risk breakdown, cash-flow projections, algorithm endpoints and OAuth2/JWT authentication. Test files, logs, commands and environment were not supplied, so we report these figures as provided and have not reproduced them.

B. What the Result Shows

A 20/20 pass shows that, in the team's environment, the assertions held. It does not establish financial accuracy, forecasting performance, anomaly-detection quality, security, scalability or production readiness, and it leaves open how strong the assertions are and whether authorization failures were tested.

C. Proposed Evaluation (Not Yet Performed)

Forecasting. Split data chronologically and use rolling-origin evaluation (Fig. 7) [9]. For n test points,

$$\mathrm{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|, \quad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}. \tag{11}$$

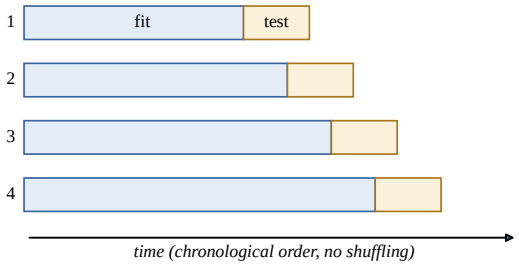


Figure 7. Proposed rolling-origin evaluation: models are fit on earlier data and tested on the following period.

Table V
RECOMMENDED ADDITIONAL TESTING

Area	Test	Evidence
Algorithm correctness	Edge-case tests per algorithm	Documented cases
Forecasting	Time-ordered MAE, RMSE vs naive	Error metrics
Anomaly detection	Precision, recall, FPR on labels	Detection metrics
API reliability	Invalid input, forged tokens, roles	Authorization behavior
Load/performance	Concurrent users, event rate	Latency, throughput
Security	Independent review, dependency audit	Findings report
Reproducibility	Published commands, seeds, logs	Re-runnable results

RMSE penalizes large errors more than MAE; percentage errors can be undefined near zero, so scaled errors are preferable across series [8]. OLS and exponential smoothing should be compared with each other and with a naive baseline on the same split.

Anomaly detection. On labeled data report

$$P = \frac{TP}{TP+FP}, \quad R = \frac{TP}{TP+FN}, \quad \mathrm{FPR} = \frac{FP}{FP+TN}. \tag{12}$$

Because anomalies are rare, precision–recall analysis is more informative than ROC alone [22]. **Risk estimation** needs calibration against observed outcomes. **Constraints and rules** should be checked against independently coded checkers and tested at thresholds and with missing values. Table V summarizes recommended additional tests.

VII. Discussion

Each method answers a different question: A* and minimax search over actions, CSP and rules check conformity, Bayesian updating revises belief, Isolation Forest surfaces outliers, and OLS and smoothing project trends. A shared workflow lets their outputs face common scrutiny. This is the design argument of FINOS, not a measured result.

Several concerns remain. Validate and verify depend on configured rules, so wrong rules let a flawed candidate pass. A deterministic result, a probability, an anomaly score and a point forecast carry different uncertainty, and showing them side by side can imply equal reliability. Simulation confirms internal consistency but not real outcomes. The replanning loop has no documented bound. The autonomy of the 11 agents is unspecified, which determines the governance burden, and any simulated data means observed behavior reflects the simulator. To support a claim of improved organization or review, a study would need a baseline workflow, approved representative data, defined measures such as time to identify an exception, and relevant users; none exists yet.

VIII. Limitations and Ethical Considerations

Limitations. (1) FINOS is an academic prototype, not a certified production system. (2) Some financial and market data may be simulated. (3) The reported tests have not been independently reproduced. (4) Forecasting and detection accuracy must be measured on suitable datasets. (5) Agent autonomy and the A* and minimax use cases must be confirmed from code. (6) Real use would need stronger security, access control, data governance, auditability and institutional approval. (7) This paper rests on a project description, not a source-code review.

Ethics. Polished dashboards invite automation bias, so outputs should show assumptions and decisions should stay with accountable humans [3]. An anomaly flag must trigger review, never automatic sanction. Skewed data can make detectors flag some groups disproportionately. Real financial records need approvals, minimization and access control; token handling and authorization need independent review; rules, parameters and model versions should be recorded for reconstruction [2]; and simulated approvals, ticks and transactions must be labeled as such.

IX. Conclusion and Future Work

FINOS integrates monitoring, budget constraints, risk estimation, anomaly detection and forecasting in one prototype whose seven algorithm families are coordinated by 11 agents through a nine-stage workflow with explicit validation, simulation, verification and audit. Its contribution is architectural: a common structure that separates generating from accepting a candidate and keeps outputs reviewable. The reported 20/20 tests do not establish accuracy, detection quality, security or readiness.

Future work includes reproducible benchmarking, a forecast comparison using MAE, RMSE and a scaled error [8], anomaly-detection evaluation with labeled data, calibration of the Bayesian estimates, explanations showing the rules fired and constraints binding, documentation of the agents and their iteration bounds, security hardening, testing with approved representative datasets, usability studies against separate dashboards, and a systematic comparison with existing platforms.

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